Coherent and Non-Coherent Receiver Structures

In digital communication systems, the receiver plays a critical role in recovering the transmitted information from the received noisy signal. Depending on whether the receiver requires carrier phase information or not, receivers are broadly classified into:

- ❖ Coherent Receivers
- ❖ Non-Coherent Receivers

The choice of receiver structure significantly affects:

1. Bit Error Rate (BER)

$$BER = \frac{No.of\ Erroneous\ Bits}{Total\ No.of\ \ Bits} = Q\sqrt{\frac{2E_b}{N_0}}\ \ [BPSK, AWGN]$$

$E_b = Bit\ Energy = Amount\ of\ Energy\ used\ to\ Transmit\ one\ bit\ of\ information$

$Q = probability\ of\ error\ caused\ by\ Gaussian\ noise.$

$N_0 = Noise\ Power\ Spectral\ Density$

2. Receiver complexity
3. Synchronization requirements
4. Power efficiency

Main components of a coherent receiver

A coherent receiver is designed to detect a transmitted signal by using a locally generated reference carrier that is synchronized in both frequency and phase with the incoming signal. Its main components work together to recover the original digital information with high accuracy.

1. **RF Front-End (Antenna + Bandpass Filter)**
   a) Receives the incoming signal $r(t)$.
   b) Filters out unwanted frequencies and noise outside the signal band.
   c) Prepares the signal for further processing.

2. **Local Oscillator (Carrier Generator)**
   a) Generates a reference carrier signal:

$$cos(2\pi f_c t + \phi)$$

   b) Must be synchronized with the transmitter in frequency and phase
   c) This is essential for coherent detection

3. **Mixer (Product Detector)**
   a) Multiplies the received signal with the local oscillator
   b) Performs frequency translation to baseband or intermediate frequency
   c) Core step in coherent demodulation

4. **Low-Pass Filter (LPF)**
   a) Removes high-frequency components created during mixing
   b) Extracts the baseband (original information-bearing signal)

5. **Sampler / Integrator**
   a) Samples the filtered signal at correct symbol intervals
   b) In many systems, performs integration over a bit period (matched filtering concept)

6. **Decision Device**
   a) Makes the final bit decision based on a threshold rule
   b) Example (BPSK):
      a. positive value → bit 1
      b. negative value → bit 0

Example

$$r(t) = s(t) + n(t)$$

Let, $r(t) = \sqrt{\frac{2E_b}{T_b}}(cos(2\pi f_c t)$

The mixer multiplies $r(t)$with a locally generated carrier:

$$cos(2\pi f_c t)$$

$$y(t) = r(t)cos(2\pi f_c t)$$

$$=> y(t) = \sqrt{\frac{2E_b}{T_b}}(cos(2\pi f_c t)(cos(2\pi f_c t))$$

A= $2\pi f_c t$

$B = 2\pi f_c t$

$$y(t) = \sqrt{\frac{2E_b}{T_b}} cosA.cosB$$

$$=> \ y(t) = \sqrt{\frac{2E_b}{T_b}} \times \frac{1}{2} \times (\cos(A - B) + \cos(A + B))$$

$$=> \ y(t) = \sqrt{\frac{2E_b}{T_b}} \times \frac{1}{2} \times (\cos(2\pi f_c t - 2\pi f_c t) + \cos(2\pi f_c t + 2\pi f_c t))$$

$$=> \ y(t) = \sqrt{\frac{2E_b}{T_b}} \times \frac{1}{2} \times (\cos 0 + \cos(4\pi f_c t)$$

$$=> y(t) = \sqrt{\frac{2E_b}{T_b}} \times \frac{1}{2}(1 + cos(4\pi f_c t))$$

$$=> y(t) = \frac{1}{2}\sqrt{\frac{2E_b}{T_b}}(1 + cos(4\pi f_c t))$$

The LPF removes the high-frequency term:

a) Removes cos $(4\pi f_c t)$

b) Keeps only the baseband component

Advantages of Coherent Receiver

a) Excellent BER performance
b) Optimal signal detection
c) High noise immunity
d) Efficient power utilization

Disadvantages Coherent Receiver

a) Complex synchronization circuitry
b) Sensitive to phase errors
c) Higher implementation cost